

Incident Lifecycle Automation in ServiceNow

PHASE 5: DEVELOPMENT

Project Detail	Information
Domain	Information Technology
Project Type	Individual Project
Platform	ServiceNow
Primary Area	IT Service Management (ITSM)
Project Scenario	Corporate VPN incident lifecycle from creation to resolution

Prepared as part of the seven-phase project documentation

1. Phase Overview

The development phase documents the ServiceNow work that was carried out to build the incident lifecycle scenario. The implementation follows a Corporate VPN issue from the initial service context through incident creation, classification, knowledge assistance, reassignment, Level 2 investigation, change management, child-incident handling, resolution, knowledge creation, and final validation.

This document deliberately focuses on the steps that are part of the implemented scenario. The broader project brief mentions additional items such as notifications and approval workflows. Those items should be demonstrated only if they were separately configured in the instance; otherwise, they remain requirements or possible enhancements rather than completed development work.

2. Environment Preparation

Development begins by confirming access to the ServiceNow environment and the roles needed to work with incident, knowledge, change, and service records. In a training or developer instance, an administrator account may be used for configuration, while impersonation can be used during testing to show how different support roles experience the same record. Sample users, groups, and configuration items can be used when real organizational data is not available.

3. Service Creation - Remote Access

The first major configuration step is to establish the service context. A service named “Remote Access” is used to represent the broader business capability that allows employees to access internal resources from outside the office. Creating or verifying this service makes the incident more meaningful because the affected business capability is visible on the record instead of being represented only by a generic network category.

- Open the appropriate ServiceNow service or service portfolio area available in the instance.
- Create or verify the service named Remote Access.
- Add an understandable description explaining that the service supports secure remote connectivity for employees.
- Save the record and confirm that it can later be selected from the incident form or related service fields.

4. Service Offering Creation - Corporate VPN

A service offering named “Corporate VPN” is created under the Remote Access service. The service offering represents the specific functionality used in the incident scenario. This distinction is useful because the service describes the broader capability while the offering identifies the exact service experience that is failing for the user.

- Create or verify the Corporate VPN service offering.
- Associate the offering with the Remote Access service.
- Use a clear description that explains that the offering provides secure VPN access to corporate resources.
- Confirm that the service offering is available when the incident is classified.

5. Incident Creation

The main incident represents an employee who cannot connect to the Corporate VPN from a remote location. The short description should make the symptom immediately understandable, for example “Unable to connect to Corporate VPN from home office” or an equivalent VPN authentication/connectivity description. The detailed description can capture the error message, when the issue started, and any basic troubleshooting already attempted.

At this stage, the goal is not to solve the problem immediately. The goal is to create a complete and traceable record that contains enough context for the Service Desk to begin analysis and, if necessary, transfer the work to a specialist team.

6. Incident Classification

The incident is then classified so that it can be understood and routed consistently. In the implemented scenario, the Category is Network, the Subcategory is VPN, and the urgency is set to Medium. The Remote Access service and Corporate VPN service offering are also used to provide business context. A suitable configuration item can be selected when the scenario requires device or infrastructure context.

Field	Project value / approach	Purpose
Category	Network	Places the issue in the correct technical area.
Subcategory	VPN	Narrows the issue to remote-access VPN support.
Urgency	Medium	Represents the urgency used in the project scenario.
Service	Remote Access	Identifies the affected business service.
Service Offering	Corporate VPN	Identifies the specific affected offering.
Configuration Item	Available project CI where applicable	Provides technical context for investigation and change work.

7. Knowledge Search and Agent Assist

Before escalating the incident, the Service Desk checks whether existing knowledge can help with the VPN issue. In a workspace that supports Agent Assist, the agent can search for relevant VPN troubleshooting content from the incident itself. If Agent Assist is not available in exactly the same layout, the standard Knowledge search experience can be used. The important development objective is to demonstrate that support knowledge is part of the troubleshooting process rather than a separate, unused repository.

A useful article can be reviewed and, where the interface supports it, associated with the incident. The agent should still apply judgment and compare the article to the user’s symptoms instead of assuming that the first search result is the correct solution.

8. Watch List and Communication Visibility

The project also uses the incident watch list to show how additional stakeholders can follow the incident. Adding relevant users to the watch list improves visibility without changing the assignment owner. In a real environment, communication behavior depends on notification configuration, so the demonstration should focus on the visible watch-list relationship unless notifications were specifically configured and tested.

9. Reassignment and Escalation to Network / Level 2

When first-line troubleshooting is not sufficient, the incident is reassigned from the Service Desk to the Network team. This handoff represents functional escalation: the incident remains the same record, but ownership moves to a team with deeper technical expertise. Before reassignment, work notes should summarize what has already been checked so that the Network team does not have to repeat the same investigation.

- Update the assignment group from Service Desk to the Network or Level 2 support group used in the instance.
- Add work notes explaining why escalation is required and what troubleshooting has already been completed.
- Save the record and verify that the assignment history remains visible.

10. Level 2 Investigation

The Network or Level 2 technician reviews the incident, service context, CI details, existing work notes, and any knowledge used by the Service Desk. During the project demonstration, impersonating or switching to the Level 2 technician role can make the handoff easier to understand. The technician then records deeper investigation findings in work notes and determines whether the issue can be fixed directly or requires a controlled change.

11. Emergency Change Creation

If the investigation shows that the VPN issue requires an urgent change to the environment, an emergency change is created and related to the incident. The exact technical content of the change should match the scenario used during the implementation, but the important project concept is the linkage between Incident Management and Change Management. The incident explains the service impact and urgency, while the change record provides a controlled place to describe and track the technical modification.

- Create an emergency change from the appropriate Change Management interface.
- Describe the reason for the change in language that connects it to the VPN incident.
- Link the change to the parent incident using the available related-record fields or related lists.
- Update the incident work notes so that the change activity is visible in the incident history.

12. Child Incident Creation

A child incident is created to represent another user or related occurrence affected by the same underlying VPN problem. The child record can have its own caller and user-facing details while remaining linked to the parent incident that contains the main investigation. This demonstrates how related incidents can be organized without losing individual user records.

During validation, both the parent and child relationship should be visible. If the parent resolution applies to the child issue, the child can be resolved with appropriate notes so that the final project state shows that both records were handled consistently.

13. Cause and Resolution Documentation

Once the technical fix has been verified, the support team records the probable cause and resolution information. Good resolution notes should explain what was wrong, what action restored service, and any important verification step. They should be understandable to another support analyst who was not involved in the original investigation. This is one of the most important quality steps because a resolved state without meaningful resolution notes provides very little value for future incidents.

14. Resolve the Parent Incident

The incident is moved to the resolved state only after the support team has enough evidence that the VPN service is working again or the agreed resolution has been delivered. Before saving the resolved state, the project should verify that the assignment, work notes, cause, resolution notes, related change, and child incident are consistent with the story being demonstrated.

15. Knowledge Article Creation

After the incident has been successfully resolved, the verified troubleshooting information is converted into a knowledge article. The article should use a clear title, describe the symptoms, provide the validated resolution steps, and avoid unnecessary environment-specific details that would confuse future readers. The purpose is to turn one successful incident resolution into reusable support knowledge.

Knowledge article section	Recommended content
Title	A concise VPN symptom and solution title.
Symptoms	What the user experiences, including connection or authentication failure.
Cause	The verified underlying reason discovered during investigation.
Resolution	The validated steps used to restore the VPN service.
Verification	How the support team confirmed that the solution worked.

16. SLA and Related-Record Validation

The final technical step is to review the incident as a complete record. The project checks the parent incident status, child incident status, linked change, knowledge output, activity history, and available SLA information. This final review is important because it proves that the project is more than a sequence of screenshots: the records remain connected and tell one consistent incident story.

17. End-to-End Test Result

Checkpoint	Expected result
Parent incident	Resolved with cause and resolution information.
Child incident	Linked to the parent and resolved as appropriate.
Assignment history	Shows the handoff from Service Desk to Network / Level 2.
Knowledge use	Relevant knowledge is reviewed during troubleshooting.
Emergency change	Exists and is visibly related to the incident when required.
Knowledge article	Created from the verified resolution.
SLA information	Visible for final lifecycle validation.

18. Development Notes and Limitations

The project brief describes a broad Incident Management automation objective that includes notifications and approval workflows. If these items were not explicitly configured during the implementation steps, they should not be presented as completed development. A strong project report is more credible when it clearly distinguishes completed features from intended enhancements. The implemented lifecycle already demonstrates substantial integration between Incident Management, Knowledge Management, Change Management, service context, related incidents, and SLA visibility.

19. Phase Conclusion

The development phase brings the design to life in ServiceNow. The Corporate VPN scenario demonstrates how a single incident can move across service context, support teams, knowledge, change activity, related incidents, resolution, and knowledge creation. The final validation confirms that the records are connected and that the lifecycle can be explained end to end during project evaluation.